Natalia Agapova

ML Engineer

natagapova.ru · Telegram · Email · Google Scholar · GitHub

Applied ML: NLP, explainability and debiasing, computer vision, RAG. I ship models with metrics, ablations, and honest trade-off analysis: accurate, explainable, and product-ready. Published research; I connect models to interfaces.

EXPERIENCE

Croissan Studio · ML, design — Apr 2025 - present

AI products in a full-cycle studio: models, interfaces, and how results are presented.

Asimov Lab · ML, UX/UI, front-end — Feb - Sep 2024

AI test-generation startup; front-end team lead. Full path from data to UI.

Freelance · ML — 2022 - present

Applied ML alongside design and web development.

SELECTED PROJECTS

Fair resume screening — NLP · debiasing · paper

BERT resume classifier for 9 IT supercategories (HeadHunter). Integrated Gradients (Captum) for bias audit; 39+ debiasing configs. Baseline: 60.9% accuracy, city-swap flipped predictions in 7.7% of pairs.

Emotion detection — CV · FER2013 · on-device

EmotionCNN: 58.7% accuracy, macro-F1 0.569; up to 60.8% with ensemble+TTA. INT8 quantization □ Core ML and ONNX Runtime Web, 1.15 ms latency.

Gesture input — CV · MediaPipe · real-time

Webcam cursor control via pinch / ready / click gestures; PyAutoGUI, smoothing, 60 FPS.

Personal Knowledge System — RAG · NLP · Python

Answers from personal PDFs grounded in context only, with filename+page citations. Chunking, SentenceTransformer, ChromaDB, Ollama llama3.2.

SKILLS

Python, PyTorch · BERT, Captum · debiasing, counterfactuals · RAG, ChromaDB, Ollama · MediaPipe, OpenCV · Core ML, ONNX

Russian C2 (native) · English C1+ · French A2 · Korean A1

EDUCATION

Innopolis University

BSc in Applied Artificial Intelligence

2026

PUBLICATIONS

Improving Fairness in AI-Powered Recruitment: An Interpretable Resume Screening System

Agapova N. A.; Lukmanov R. A., 2026

doi:10.66693/mathai.1017

Quantum Annealing Approaches to Breaking RSA Encryption

Kholodov Y. A.; Salloum H.; Agapova N. A. *Voprosy kiberbezopasnosti*, 2025

doi:10.21681/2311-3456-2025-4-127-133